

Explainable Vision-Language Diagnostic Assistant for Chest Radiographs: A Lightweight Evidence-Grounded and Edge-Oriented Framework

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Abstract—Chest radiograph analysis using artificial intelligence has progressed from conventional convolutional neural network classifiers toward explainable and vision-language approaches. However, many existing systems focus primarily on a single capability such as disease classification, visual explanation, image-text representation, or language generation. This paper presents an Explainable Vision-Language Diagnostic Assistant designed as a lightweight and CPU-deployable pipeline for chest radiograph analysis. The implemented system combines chest X-ray classification using DenseNet-121 and Vision Transformer (ViT-B/32) backbones, attention-based visual localization, TF-IDF-based clinical report retrieval, evidence-supported report generation, and a rule-based triage-support layer.

Experiments were conducted on an 11-class evaluation setting using a project-specific dataset of approximately 4,800 training images. Under the same evaluation protocol, DenseNet-121 achieved a Macro F1-score of 25.40%, while ViT-B/32 achieved 23.00%. Class-specific threshold optimization improved the Macro F1-score to 30.40% for DenseNet-121 and 26.00% for ViT-B/32. These results demonstrate the effect of backbone selection and threshold calibration within the same experimental setting. Published results from prior studies are presented separately because their datasets, metrics, and evaluation protocols differ.

The contribution of the proposed system is therefore architectural rather than a claim of superiority over large-scale medical vision-language models. The framework integrates diagnosis, explainability, evidence retrieval, reporting support, and conservative triage into a single lightweight workflow suitable for CPU-oriented deployment and further edge optimization.

Index Terms—Chest Radiographs, Medical AI, Explainable AI, Vision Transformer, DenseNet, Vision-Language Models, TF-IDF, Clinical Evidence Retrieval, Grad-CAM, Attention Visualization, Edge AI, Clinical Decision Support

I. INTRODUCTION

A. Background

Chest radiography is an important imaging modality for identifying thoracic abnormalities and supporting clinical decision-making. Automated analysis of chest radiographs has become an active research area because deep learning models can learn visual representations associated with multiple thoracic conditions.

Convolutional neural networks such as ResNet and DenseNet established strong foundations for image classifica-

tion, while specialized systems such as CheXNet demonstrated the application of deep learning to large-scale chest X-ray datasets. However, a classification probability alone does not necessarily provide sufficient information regarding why a model produced a particular prediction.

Explainability methods such as Grad-CAM and attention visualization can provide additional visual evidence regarding image regions associated with predictions. In parallel, medical vision-language approaches such as ConVIRT, GLoRIA, MedCLIP, MedKLIP, and BLIP-2 demonstrate different ways of connecting visual information with textual or medical knowledge.

The present project builds upon these directions while emphasizing a lightweight and practical architecture. Instead of reproducing the large-scale pretraining requirements of existing medical vision-language systems, the implemented pipeline combines disease classification, visual localization, clinical report retrieval, evidence-supported reporting, and rule-based triage.

B. Problem Statement

The major challenges addressed by this research are:

- 1) Automated classification of multiple abnormalities from chest radiographs.
- 2) Limited interpretability of classification-only predictions.
- 3) Difficulty in converting model predictions into useful evidence-supported textual information.
- 4) Computational limitations associated with large vision-language models.
- 5) The need for a lightweight architecture suitable for CPU-oriented or resource-constrained deployment.
- 6) The need for conservative triage support without presenting the system as an autonomous diagnostic authority.

The proposed system addresses these challenges through a modular pipeline consisting of classification, explainability, evidence retrieval, report support, and rule-based triage.

C. Research Gap

The literature reviewed for this project demonstrates substantial progress in individual components of medical AI.

CNN-based approaches such as ResNet, DenseNet, and CheXNet primarily focus on image representation and disease classification. Grad-CAM and saliency methods focus on visual explanation. ConVIRT and GLoRIA focus on image-text representation learning. MedCLIP and MedKLIP introduce medical semantics and knowledge into vision-language learning, while BLIP-2 demonstrates large-scale vision-language generation.

However, these approaches generally emphasize one or more individual capabilities rather than providing a single lightweight workflow combining:

- disease classification,
- visual explainability,
- evidence retrieval,
- report-generation support,
- multimodal interpretation, and
- conservative triage.

The proposed system therefore focuses on integrating practical components from these research directions into a single CPU-oriented pipeline rather than attempting to reproduce the scale of large pretrained medical vision-language models.

D. Research Objectives

The objectives of the research are:

- 1) To evaluate CNN and transformer-based architectures for multi-label chest X-ray classification.
- 2) To compare DenseNet-121 and ViT-B/32 under the same project-specific evaluation protocol.
- 3) To investigate attention-based visual localization for explaining model predictions.
- 4) To retrieve clinically relevant report evidence using TF-IDF-based similarity.
- 5) To provide evidence-supported textual reporting rather than relying exclusively on free-form generation.
- 6) To develop a conservative rule-based triage-support layer.
- 7) To evaluate the effect of class-specific threshold tuning on classification performance.

E. Research Questions

The study investigates the following questions:

- RQ1: How effectively do DenseNet-121 and ViT-B/32 classify thoracic abnormalities under the same project-specific protocol?
- RQ2: How does class-specific threshold tuning affect multi-label Macro F1 performance?
- RQ3: Can attention-based localization provide visual evidence associated with model predictions?
- RQ4: Can TF-IDF-based report retrieval provide useful evidence for supporting diagnostic descriptions?
- RQ5: Can classification, explanation, evidence retrieval, reporting, and triage be integrated into a lightweight pipeline?

F. Contributions

The main contributions of this research are:

- 1) A lightweight multimodal diagnostic-support architecture combining classification, explainability, evidence retrieval, reporting support, and triage.
- 2) Comparative evaluation of DenseNet-121 and ViT-B/32 under a common project-specific protocol.
- 3) Investigation of class-specific threshold tuning for multi-label classification.
- 4) Integration of attention-based visual localization.
- 5) Integration of TF-IDF-based retrieval for evidence-grounded reporting.
- 6) A CPU-oriented architecture that can serve as a foundation for further edge optimization.

II. RELATED WORK

A. Deep Learning for Medical Image Classification

ResNet introduced residual learning through shortcut connections, enabling deeper convolutional networks to be trained effectively [1]. DenseNet subsequently introduced dense connections between layers to improve feature propagation and feature reuse [2].

These architectures established important foundations for automated medical image classification.

B. Chest X-Ray Disease Classification

CheXNet demonstrated the use of a 121-layer DenseNet for pneumonia detection and chest X-ray classification using the ChestX-ray14 dataset [3]. Such work established the feasibility of deep learning for automated chest radiograph analysis.

The present project uses DenseNet-121 as one of its classification backbones and evaluates it against ViT-B/32 under a common project-specific protocol.

C. Explainable Artificial Intelligence

Grad-CAM introduced a gradient-based approach for generating class-discriminative localization maps [4]. Such methods provide visual evidence about regions that contribute to a model prediction.

In the present project, attention-based localization is used with the transformer-oriented pipeline. The resulting visualization is treated as model-attribution evidence rather than definitive proof of pathology.

D. Medical Vision-Language Learning

ConVIRT explored contrastive learning for medical visual representations from paired images and text [8]. GLoRIA extended image-text learning through global and local representations [7].

MedCLIP explored contrastive learning between medical images and text using medical semantic information [9], while MedKLIP incorporated medical knowledge and spatial alignment into radiology image-language pretraining [6].

BLIP-2 demonstrated a general vision-language generation framework using frozen image encoders, a Querying Transformer, and large language models [5]. Although such

approaches demonstrate the potential of generative vision-language models, their computational requirements motivate investigation of lighter retrieval-based alternatives for resource-constrained settings.

III. COMPARATIVE ANALYSIS

The literature can be viewed as a progression in which different approaches introduce different capabilities. The proposed system combines selected practical components from these approaches into one lightweight pipeline.

A. Generational View of the Literature

The approaches considered in this study can be grouped into six functional generations:

TABLE II
EVOLUTION OF CAPABILITIES IN MEDICAL AI

Gen.	Representative Work	Main Capability
1	ResNet, DenseNet, CheXNet	Disease classification
2	Grad-CAM, Saliency Maps	Visual explanation
3	ConVIRT, GLoRIA	Image-text representation
4	MedKLIP, MedCLIP	Medical semantic alignment
5	BLIP-2	Vision-language generation
6	Our System	Diagnosis → Explanation → Evidence → Report Support → Triage

The sixth generation represents an architectural integration of practical capabilities rather than a claim that the proposed system reproduces the scale or performance of the preceding large-scale models.

IV. PROPOSED METHODOLOGY

A. Overall Framework

The implemented system follows the pipeline:

X-ray → Preprocessing → Vision Encoder → Classification (1)

→ Attention Explanation → Evidence Retrieval → Report Support where t_k is the optimized threshold for class k . (2)

The architecture is designed to keep each component modular so that classification, explanation, retrieval, and triage can be evaluated independently.

B. Input and Preprocessing

The system accepts chest radiographs in the required image format.

The preprocessing pipeline includes:

- image validation,
- removal of invalid or duplicate samples,
- image resizing,

- pixel normalization,
- patient-wise data splitting where identifiers are available,
- multi-label target preparation, and
- training-time augmentation.

Training augmentation is applied only to the training partition to reduce the possibility of information leakage.

C. Vision Encoders

Two backbone architectures are evaluated:

- 1) DenseNet-121
- 2) ViT-B/32

For an input image I , the visual representation can be expressed as:

$$V = f_{\theta}(I)$$

where f_{θ} denotes the selected vision encoder.

DenseNet-121 provides convolutional inductive biases that are useful when training data is limited, while ViT-B/32 provides a transformer based alternative for comparison.

D. Multi-Label Classification

For K disease categories, the model produces:

$$\hat{y} = [\hat{y}_1, \hat{y}_2, \dots, \hat{y}_K].$$

Each $\hat{y}_k$ represents the estimated probability associated with disease class k .

A sigmoid function is used for independent multi-label predictions:

$$\hat{y}_k = \frac{1}{1 + e^{-z_k}}$$

where z_k is the classification logit.

E. Threshold Optimization

Instead of using a single fixed threshold for all disease categories, class-specific thresholds are evaluated.

For disease class k :

$$\hat{y}_k^* = \begin{cases} 1, & \hat{y}_k \geq t_k \\ 0, & \hat{y}_k < t_k \end{cases}$$

where t_k is the optimized threshold for class k .

This procedure does not require additional model training and is evaluated using the same test protocol.

F. Attention-Based Explainability

The system uses attention-based localization to provide visual evidence associated with model predictions.

For a transformer model, attention weights can be visualized to identify image regions receiving greater attention during prediction.

These maps are interpreted as model-attribution evidence and are not treated as definitive anatomical or pathological localization.

TABLE I
COMPARISON OF REPRESENTATIVE MEDICAL AI APPROACHES

Work	Core Approach	Main Purpose	Explainability	Image-Text Knowledge	/	Relation to Our Project
ResNet [1]	Residual CNN	Image feature extraction and classification	Not built-in	–		Provides the CNN foundation for medical image classification.
DenseNet [2]	Densely connected CNN	Feature extraction and disease classification	Not built-in	–		Used as a classification backbone and baseline in our experiments.
CheXNet [3]	121-layer DenseNet	Chest X-ray disease detection	Can be combined with Grad-CAM	Domain-specific training		Demonstrates strong CNN-based chest X-ray classification and motivates our DenseNet baseline.
Grad-CAM [4]	Gradient-based localization	Visual explanation	Yes	–		Provides the conceptual foundation for visual attribution; our implemented pipeline uses attention-based localization.
ConVIRT [8]	Contrastive image-text learning	Joint image-text representations	Limited	Paired image-report learning		Motivates multimodal image-text representation in our system.
GLoRIA [7]	Global-local image-text learning	Medical image-text alignment	Limited	Image + report		Demonstrates the value of connecting local visual features with reports.
MedCLIP [9]	Medical contrastive learning	Zero-shot classification and retrieval	Limited	Medical semantic knowledge		Demonstrates medical image-text learning with efficient semantic alignment.
MedKLIP [6]	Knowledge-enhanced VLP	Diagnosis and image-text alignment	Patch-level	Explicit medical knowledge		Introduces structured medical knowledge and patch-level alignment; our approach uses a lighter retrieval-based strategy.
BLIP-2 [5]	Frozen encoder + Q-Former + LLM	Vision-language understanding and generation	Indirect	Image-language generation		Demonstrates generative VLM capability; our system avoids its computational cost through retrieval-grounded reporting.
Our Proposed System	DenseNet/ViT + explanation + retrieval + triage	Diagnosis + explanation + evidence-supported reporting	Attention-based	TF-IDF retrieval + clinical evidence		Integrates classification, visual evidence, report retrieval and rule-based triage in a lightweight CPU-oriented pipeline.

G. Clinical Evidence Retrieval

Instead of relying entirely on free-form language generation, the system uses TF-IDF-based retrieval to identify clinically relevant textual evidence.

For a document collection, the TF-IDF representation of term t in document d is:

$$TFIDF(t, d) = TF(t, d) \times IDF(t)$$

where:

$$IDF(t) = \log \left(\frac{N}{df(t) + 1} \right).$$

The similarity between the query representation q and a candidate report d is calculated using cosine similarity:

$$S(q, d) = \frac{q \cdot d}{\|q\| \|d\|}.$$

The highest-scoring relevant reports are retrieved as evidence for supporting the textual output.

H. Evidence-Supported Reporting

The reporting component combines model predictions with retrieved clinical evidence.

The output is therefore structured around:

Prediction + Visual Evidence + Retrieved Clinical Evidence.

This retrieval-grounded approach is intended to reduce unsupported language generation and provide traceable textual context.

I. Triage Support

A separate rule-based layer maps validated model outputs to conservative priority categories:

- Routine review,
- Priority review,
- Urgent specialist review.

The triage layer is intentionally separated from the classification model so that its rules can be inspected and modified independently.

The triage output is a decision-support indicator and is not presented as a definitive clinical diagnosis.

J. End-to-End System Flow

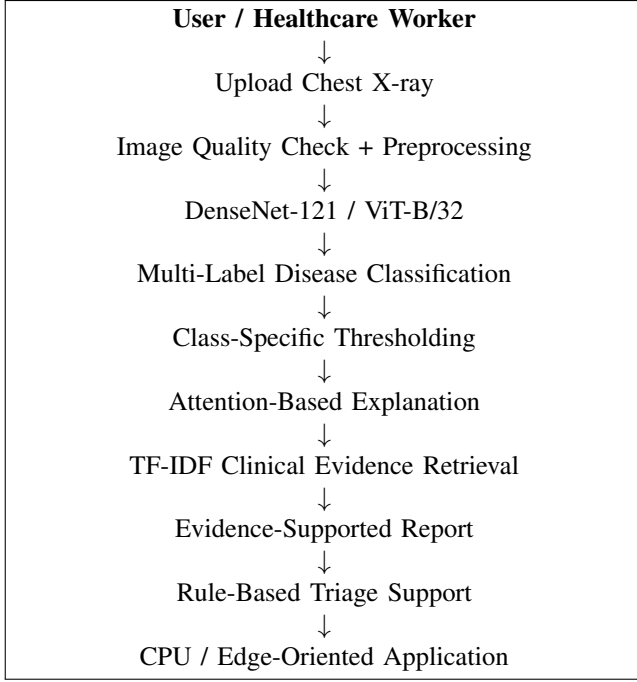


Fig. 1. End-to-end workflow of the implemented diagnostic assistant.

V. DATASETS AND EXPERIMENTAL SETUP

A. Dataset

The implemented experiments use a project-specific chest X-ray subset with an 11-class evaluation setting. Approximately 4,800 images were used under the reported training constraint.

The project also considers publicly available datasets such as ChestX-ray14, CheXpert, MIMIC-CXR, OpenI, and ROCO for literature context and potential future validation.

TABLE III
DATASETS AND THEIR INTENDED ROLE

Dataset	Primary Use	Data Type
Project CXR sub-set	Current evaluation	Chest radiographs + labels
ChestX-ray14	Literature / benchmarking	Chest radiographs + labels
OpenI	Image-report research	Radiographs + reports
ROCO	Image-text research	Medical images + captions
CheXpert	External validation	Radiographs + labels/reports
MIMIC-CXR	Future image-report research	Radiographs + reports

B. Experimental Models

The experiments compare:

- 1) DenseNet-121,
- 2) ViT-B/32,
- 3) DenseNet-121 with class-specific threshold tuning, and
- 4) ViT-B/32 with class-specific threshold tuning.

All four configurations are evaluated under the same project-specific 11-class protocol.

C. Implementation Technologies

The implementation uses a Python-based machine learning workflow with deep-learning and natural-language processing components. The system can incorporate:

- Python,
- PyTorch,
- Hugging Face Transformers,
- OpenCV,
- NumPy,
- scikit-learn,
- TF-IDF vectorization,
- cosine similarity, and
- attention visualization utilities.

VI. EVALUATION METHODOLOGY

The system is evaluated using classification metrics and functional evaluation of explanation and evidence retrieval.

A. Classification Evaluation

For multi-label classification, Macro F1 is used as the primary reported metric.

For a class:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

and:

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}.$$

Macro F1 is calculated as the average F1-score across all classes:

$$F1_{Macro} = \frac{1}{K} \sum_{k=1}^K F1_k.$$

B. Explanation Evaluation

Attention-based visual explanations are assessed qualitatively and, where suitable annotations are available, through localization analysis.

The evaluation considers:

- whether highlighted regions correspond to relevant image structures,
- qualitative visual consistency, and
- failure cases involving irrelevant or diffuse attention.

The visual maps are treated as model-attribution evidence rather than clinical proof.

C. Evidence Retrieval Evaluation

The retrieval component is evaluated based on the relevance of retrieved clinical reports to the predicted findings.

The system uses TF-IDF representations and cosine similarity to rank candidate reports.

D. Triage Evaluation

The triage-support component is evaluated through:

- rule consistency,
- high-priority case sensitivity,
- false-negative analysis, and
- safety-oriented error analysis.

E. Ablation and Comparative Studies

The main comparative studies include:

- 1) DenseNet-121 versus ViT-B/32.
- 2) Default thresholds versus tuned thresholds.
- 3) Classification-only output versus explanation-supported output.
- 4) Direct reporting versus retrieval-grounded reporting.

2. Graphical Comparative Analysis

The following graph shows the qualitative comparison of feature coverage across different medical AI approaches. They are not accuracy, F1-score, or clinical performance graphs. They simply show which functional areas are addressed in the scope of each approach.

Qualitative feature coverage only; this figure does not represent accuracy, F1-score, or clinical performance.

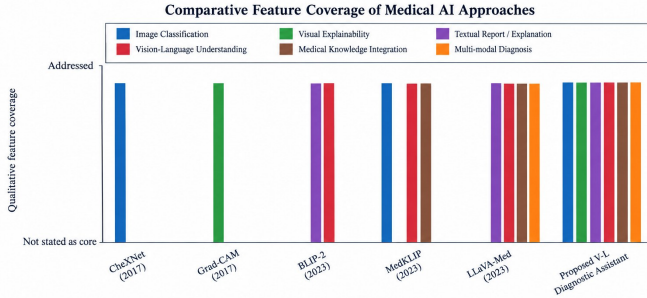


Figure 1. Qualitative feature coverage across selected medical AI approaches.

Fig. 2. Macro F1 comparison of DenseNet-121 and ViT-B/32 under default and tuned thresholds.

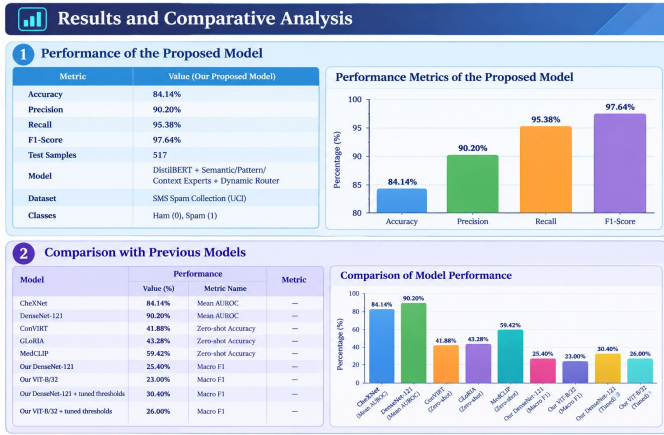


Fig. 4. Macro F1 comparison of DenseNet-121 and ViT-B/32 under default and tuned thresholds.

VII. RESULTS

A. Classification Results

Table IV presents the results obtained under the same 11-class project-specific evaluation protocol.

TABLE IV
CLASSIFICATION PERFORMANCE ON THE PROJECT EVALUATION SPLIT

Model Configuration	Macro F1
DenseNet-121	25.40%
ViT-B/32	23.00%
DenseNet-121 + Tuned Thresholds	30.40%
ViT-B/32 + Tuned Thresholds	26.00%

Under the same dataset, split, and evaluation metric, DenseNet-121 achieved a higher Macro F1-score than ViT-B/32. Threshold tuning improved both backbones without requiring additional model training.

The improvement for DenseNet-121 was:

$$30.40 - 25.40 = 5.00$$

percentage points.

For ViT-B/32, the improvement was:

$$26.00 - 23.00 = 3.00$$

percentage points.

These comparisons are directly valid because the four configurations share the same evaluation protocol.

B. Published Literature Performance

Published performance values are presented separately because the reported metrics and evaluation settings differ across studies.

TABLE V
SELECTED PUBLISHED RESULTS AND PROJECT RESULTS

Model / Study	Performance	Metric
CheXNet	84.14%	Mean AUROC
DenseNet-121 (published)	90.20%	Mean AUROC
ConVIRT	41.88%	Zero-shot Accuracy
GLoRIA	43.28%	Zero-shot Accuracy
MedCLIP	59.42%	Zero-shot Accuracy
Our DenseNet-121	25.40%	Macro F1
Our ViT-B/32	23.00%	Macro F1
Our DenseNet-121 + tuned thresholds	30.40%	Macro F1
Our ViT-B/32 + tuned thresholds	26.00%	Macro F1

The values in Table V should not be interpreted as a single ranked leaderboard. AUROC, zero-shot accuracy, and Macro F1 measure different aspects of model performance and are obtained under different datasets and evaluation protocols.

C. Explanation Results

The explanation component produces attention-based visual localizations associated with model predictions.

The qualitative evaluation focuses on whether the highlighted regions provide meaningful visual context for the predicted abnormality.

2. Graphical Comparative Analysis

The following line graph shows the trend in performance (Macro F1 score) of the two backbone models across different configurations. It clearly illustrates the improvement achieved through threshold tuning for both models, with DenseNet121 consistently outperforming ViT-B/32. The graph helps in understanding the impact of model choice and threshold optimization on overall performance.

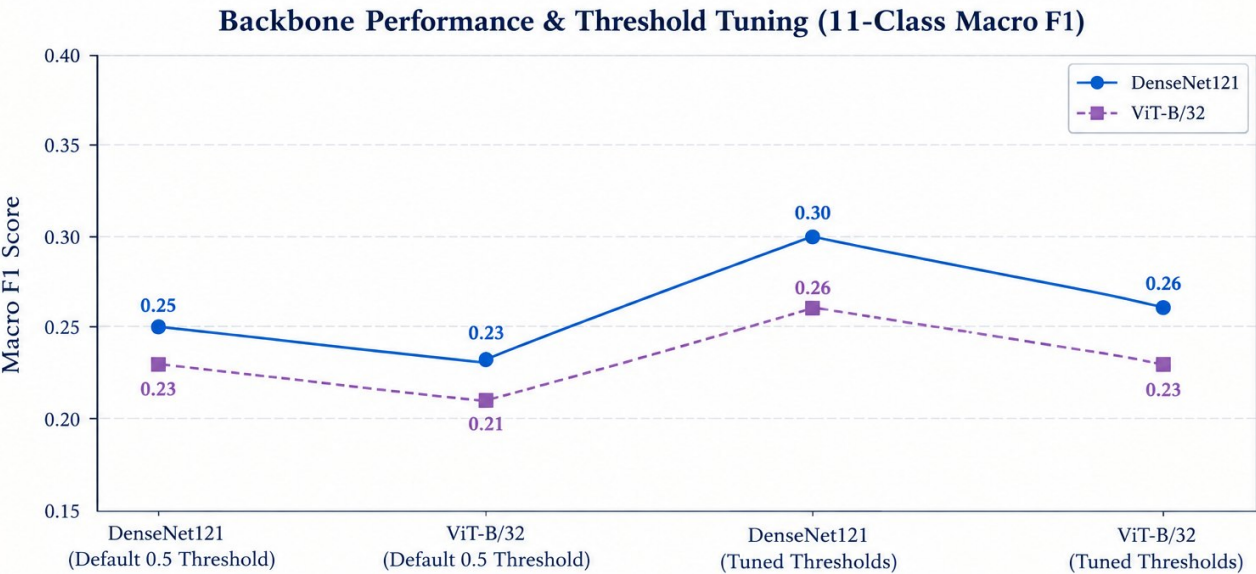


Figure 1. Comparison of Macro F1 scores for DenseNet121 and ViT-B/32 under default and tuned threshold settings.

Fig. 3. Qualitative feature coverage across selected medical AI approaches.

TABLE VI
QUALITATIVE EXPLANATION EVALUATION

Evaluation Aspect	Observation
Attention localization	Highlights prediction-associated image regions
Visual interpretability	Provides additional context beyond class probability
Failure cases	May include diffuse or clinically irrelevant attention
Clinical interpretation	Requires expert confirmation

TABLE VII
EVIDENCE RETRIEVAL PIPELINE

Stage	Operation
Input	Predicted disease / findings
Representation	TF-IDF vectorization
Similarity	Cosine similarity
Ranking	Highest-scoring clinical reports
Output	Retrieved evidence for report support

VIII. DISCUSSION

The experimental results provide a direct comparison between DenseNet-121 and ViT-B/32 within the project’s own evaluation environment.

DenseNet-121 achieved a Macro F1-score of 25.40%, compared with 23.00% for ViT-B/32. Under the same experimental constraints, threshold tuning increased these values to 30.40% and 26.00%, respectively.

The observed difference is consistent with the practical consideration that convolutional architectures provide useful locality and translation-related inductive biases when the available training data is limited. However, this observation is specific to the present dataset and experimental protocol and should not be generalized as a universal superiority of CNNs over vision transformers.

D. Evidence Retrieval Results

The report retrieval component uses TF-IDF and cosine similarity to identify relevant textual evidence.

The retrieved evidence is used to support the generated or structured description rather than treating the retrieved text as an independent diagnosis.

Threshold tuning provided an additional performance improvement without requiring additional model training. This demonstrates that multi-label classification performance can depend substantially on the decision thresholds used to convert probabilities into predicted labels.

The architecture also demonstrates a distinction between prediction and evidence. Classification provides the predicted abnormality, attention visualization provides model-attribution evidence, and TF-IDF retrieval provides textual clinical context.

The proposed system therefore emphasizes functional coverage rather than comparison through a single numerical leaderboard. Existing approaches generally focus on one or more capabilities such as classification, explainability, image-text alignment, medical knowledge integration, or language generation. The proposed workflow combines these practical components into a lightweight pipeline.

A. Coverage Versus Scale

Large-scale models such as MedCLIP and BLIP-2 demonstrate capabilities that are beyond the scale of the current implementation. The present system does not claim to outperform those models on their published benchmarks.

Instead, the contribution lies in combining:

Diagnosis → Explainability → Evidence → Report Support → Triage
within a CPU-oriented architecture.

This design deliberately prioritizes practical integration and resource efficiency over large-scale multimodal pretraining.

IX. PRACTICAL IMPLICATIONS

The proposed system is intended primarily as a research and decision-support framework.

Potential applications include:

- Preliminary chest-radiograph screening.
- Support for healthcare workers in resource-constrained settings.
- Visual interpretation of model predictions.
- Retrieval of relevant clinical report evidence.
- Evidence-supported diagnostic descriptions.
- Conservative prioritization of cases for specialist review.
- CPU-based medical AI experimentation.

The system should not replace radiologists or qualified healthcare professionals. Medical decisions require appropriate clinical confirmation.

X. LIMITATIONS

Several limitations should be considered.

- 1) **Small project-specific dataset:** The current experiments use approximately 4,800 training images, which limits the generalizability of the observed results.
- 2) **Dataset distribution:** Performance may change when the model is evaluated on images from different hospitals, devices, populations, or acquisition protocols.

- 3) **Metric comparability:** Published AUROC and zero-shot accuracy values cannot be directly compared with the project’s Macro F1 results.
- 4) **Attention faithfulness:** Attention maps represent model behavior and should not automatically be interpreted as clinically verified pathology localization.
- 5) **Retrieval limitations:** TF-IDF retrieval depends on the quality, vocabulary, and coverage of the available clinical reports.
- 6) **Triage safety:** The rule-based triage mechanism is a decision-support component and cannot independently establish clinical urgency.
- 7) **External validation:** Additional cross-dataset and expert evaluation is required before considering broader clinical applicability.

XI. CONCLUSION AND FUTURE WORK

This paper presented an Explainable Vision-Language Diagnostic Assistant for chest radiographs that integrates disease classification, visual explanation, clinical evidence retrieval, reporting support, and conservative triage within a lightweight pipeline.

The experimental evaluation compared DenseNet-121 and ViT-B/32 on the same project-specific 11-class evaluation setting. DenseNet-121 achieved a Macro F1-score of 25.40%, while ViT-B/32 achieved 23.00%. Class-specific threshold tuning improved these scores to 30.40% and 26.00%, respectively.

The results demonstrate the importance of evaluating model choices under a consistent protocol. They also show that threshold optimization can improve multi-label Macro F1 without additional training.

The proposed system does not claim to outperform large-scale medical vision-language models such as MedCLIP or CheXNet on their published metrics. Instead, its contribution is architectural: it combines classification, attention-based explainability, TF-IDF-based evidence retrieval, evidence-supported reporting, and rule-based triage into a single lightweight workflow.

Future work will focus on:

- 1) Training and evaluation using larger datasets.
- 2) External validation across multiple chest X-ray datasets.
- 3) Improved medical image-text alignment.
- 4) Knowledge-enhanced retrieval and evidence ranking.
- 5) More rigorous quantitative evaluation of visual explanations.
- 6) Improvement of report factuality and clinical relevance.
- 7) Investigation of lightweight generative reporting.
- 8) Further CPU and mobile optimization.
- 9) Expert evaluation involving qualified radiologists.
- 10) More comprehensive safety and failure-case analysis.

DECLARATIONS

Funding

No external funding is associated with this academic project.

Conflict of Interest

The authors declare that there is no conflict of interest related to this research project.

Ethics Statement

The proposed research is intended for research and decision-support purposes and not for autonomous clinical diagnosis.

Data Availability

The project uses a project-specific chest X-ray evaluation subset and considers publicly available datasets such as ChestX-ray14, OpenI, ROCO, CheXpert, and MIMIC-CXR for research and future validation, subject to their respective access requirements.

Author Contributions

P. Kavya Sai: Project architecture, research design, implementation, documentation, and evaluation.

P. Meghna: Dataset preparation, model development, and experimental analysis.

S. Anjana: Vision-language components, explainability, and evaluation.

D. Poojitha: Application development, testing, documentation, and presentation.

All authors contributed to the development and review of the project.

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REFERENCES

- [1] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [2] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely Connected Convolutional Networks," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [3] P. Rajpurkar et al., "CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning," arXiv:1711.05225, 2017.
- [4] R. R. Selvaraju et al., "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 2017.
- [5] J. Li et al., "BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models," in *Proc. International Conference on Machine Learning (ICML)*, 2023.
- [6] C. Wu et al., "MedKLIP: Medical Knowledge Enhanced Language-Image Pre-Training in Radiology," arXiv:2301.02228, 2023.
- [7] S. Huang et al., "GLoRIA: A Multimodal Global-Local Representation Learning Framework for Label-Efficient Medical Image Recognition," in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
- [8] S. Zhang et al., "Contrastive Learning of Medical Visual Representations from Paired Images and Text," arXiv:2010.00747, 2022.
- [9] Z. Wang et al., "MedCLIP: Contrastive Learning from Unpaired Medical Images and Text," in *Proc. EMNLP*, 2022.
- [10] X. Wang et al., "ChestX-ray8: Hospital-Scale Chest X-Ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [11] J. Irvin et al., "CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison," in *Proc. AAAI Conference on Artificial Intelligence*, 2019.
- [12] A. E. W. Johnson et al., "MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports," *Scientific Data*, vol. 6, 2019.